

Salazar Roofing OS

DevOps Demo Build Package

CRM + ClaimDesk + CrewDesk + Customer Portal + StormLoop + Referrals

Prepared for: DevOps / demo build team

Date: June 16, 2026

Version: 1.0-demo-build

North star

One connected flow from storm event to lead, inspection, claim-support file, job production, customer update, payment, warranty, review, referral, and repeat revenue.

Deliverables in this handoff: this PDF build brief and a structured JSON design spec named salazar_roofing_os_demo_spec.json.

1. Executive summary

This package consolidates the uploaded schema artifact, working UI prototype, crew-manager role matrix, roofing/remodeling needs breakdown, and the upgraded StormLoop/referral strategy into one demo-build handoff.

The product should be positioned as a roofing/remodeling operating system, not only a CRM. The demo should prove one connected business flow: storm event -> affected ZIPs -> canvassing mission -> lead -> inspection -> claim-support file -> production -> crew update -> customer portal -> invoice/warranty -> referral.

The highest-value differentiator is StormLoop: weather-driven growth intelligence that alerts marketing staff to major weather events, identifies affected marketing ZIP/ZCTA areas, launches ethical canvassing missions, tracks door outcomes, and ties every lead/revenue dollar back to storm, ZIP, rep, route, and referral source.

Demo goal: build a clickable, data-backed demo that makes Salazar believe the app can replace spreadsheets, texts, paper claim folders, scattered photos, manual customer updates, and manual referral tracking.

1.1 Source material reviewed

File	Lines	Purpose
Pasted text (2).txt	429	Working HTML prototype/demo: dashboard, jobs board, job detail, insurance tab, crew view, notification history, customer list, and reports.
Pasted text (3).txt	196	Crew manager role model and permission matrix artifact.
Pasted text (4).txt	277	Operational needs breakdown, roofing/remodeling software landscape, feature gap analysis, insurance workflow, and customer notification strategy.
Pasted text.txt	610	Original interactive database schema artifact: core CRM, insurance, operations, crew, communications, billing, relationships, and tech stack.

2. Product positioning

Working name: Salazar Roofing OS.

Demo tagline: Normal CRMs manage jobs after you get them. This system helps find storm opportunities, win the job, run the job, update the customer, collect the money, and turn the customer into future referrals.

Core promise: One app to find storm opportunities, capture leads, document inspections, organize insurance-support paperwork, manage crews, update homeowners, track payments, and turn completed customers into referrals.

2.1 What makes this different

- ClaimDesk treats insurance-support documentation as a first-class workflow without crossing into public-adjuster language.
- Customer portal uses secure magic links, so homeowners do not need to download an app.
- CrewDesk gives field teams structured daily logs, safety checklists, material requests, and progress photos.
- StormLoop turns weather into measurable marketing missions with affected ZIPs, routes, door outcomes, and revenue attribution.
- Referral Rewards gives customers long-term value and tracks rewards from referral link to paid job.

2.2 MVP non-goals

- Do not rebuild Xactimate or claim-adjusting tools.
- Do not process real insurance claims in demo.
- Do not send production marketing SMS before A2P 10DLC approval and consent flows are complete.
- Do not claim a specific home has storm damage without an inspection.
- Do not build direct supplier ordering before material request workflow is proven.

- Do not build native iOS/Android for demo unless responsive web/PWA fails field use case.

3. Roofing/remodeling company needs

Need area	What the app must solve	Demo proof
Leads and CRM	Capture homeowner, property, service type, source, rep, follow-up, referral/storm attribution.	Lead inbox, quick lead, jobs board.
Inspection	Mobile checklist, tagged photos, measurement/report uploads, inspection summary.	Inspection flow and photo tags on job detail.
Insurance support	Claim number, carrier, adjuster, date of loss, scope, estimate versions, supplements, ACV/RCV/deductible.	Maria Chen ClaimDesk tab with payment tracker.
Production	Move jobs from approved to materials, scheduled, crew assigned, in progress, QC, invoice, warranty.	Production board and status changes.
Crew	Safety checklist, clock-in/out, daily logs, progress photos, material requests, certifications.	CrewDesk dashboard and Crew A daily log.
Customer communication	Milestone SMS/email/portal updates, documents, invoice, warranty, messages.	Magic-link customer portal preview and notification history.
Storm response	Weather alerts, affected ZIPs, customer impact list, canvassing missions, door outcomes.	StormLoop event and mission builder.
Referrals	Referral link/QR, statuses, reward approval, loyalty tiers, top referrers.	Referral wallet and reward queue.
Remodeling	Selections, allowances, permits, change orders, punch lists, client approvals.	Remodeling board as Phase 2/3 demo extension.

4. Demo MVP scope

4.1 Modules to include in the demo

Module	MVP	Primary users	Must show in demo
CRM Command Center	Yes	owner_admin, office_manager, sales_rep_estimator	Lead inbox, Customer profile, Property profile, Sales pipeline
Inspection App	Yes	sales_rep_estimator, office_manager	Inspection checklist, Photo capture and tagging, Damage annotations, Measurement report upload
ClaimDesk Insurance-Support Module	Yes	office_manager, insurance_coordinator, owner_admin	Carrier and claim number, Date of loss, Adjuster contact, Adjuster appointment
Production Board	Yes	production_manager, office_manager, owner_admin	Roofing board, Remodeling board, Job status changes, Material readiness
CrewDesk Operations	Yes	crew_manager, crew_lead, crew_member	Assigned crew schedule, GPS clock-in/out, Safety checklist, Daily job log
Customer Portal	Yes	customer, office_manager, sales_rep_estimator	Magic link access, Project timeline, Current status, Next step
Notification Engine	Yes	office_manager, owner_admin, marketing_manager	Transactional templates, Marketing templates, SMS consent capture, STOP/HELP handling
StormLoop Growth Engine	Yes	marketing_manager, canvasser, sales_rep_estimator	Weather alert ingest, Affected ZIP/ZCTA impact scoring, Past customer impact list, Storm mission builder

Module	MVP	Primary users	Must show in demo
Referral Rewards Engine	Yes	customer, marketing_manager, office_manager	Customer referral link, Referral QR code, Referral status tracking, Reward approval workflow

4.2 Recommended demo stack

Layer	Recommendation
frontend	Next.js + TypeScript + Tailwind/shadcn or equivalent component library
backend	Next.js API routes or NestJS/Node API for demo
database	PostgreSQL/Supabase with row-level security design
storage	S3-compatible storage or local mock storage for demo files/photos
maps	Mapbox or Google Maps for StormLoop route mock; can start with static map/pins
messaging	Mock SMS/email provider in demo; provider adapter for Twilio/SendGrid later
weather	Simulated storm endpoint first; NWS Alerts API worker adapter behind feature flag
deployment	Vercel/Render/Fly.io for demo app; Supabase for database; seed script included

4.3 Suggested repo structure

- /apps/web - Next.js dashboard and portal
- /apps/api - API service or route handlers
- /packages/ui - shared UI components
- /packages/shared - types, enums, validation schemas
- /packages/db - migrations, RLS policies, seed scripts
- /packages/integrations - weather, SMS, email, accounting adapters
- /docs - demo script, API spec, screenshots

5. System architecture

StormLoop -> CRM -> Inspection App -> ClaimDesk -> Production Board -> CrewDesk -> Customer Portal -> Finance/HR

Data hub: jobs table

Security: company_id tenant isolation + role-based access + row-level security

External adapters: weather, SMS/email, file storage, QuickBooks, measurement reports, photo providers, suppliers

Demo mode: seeded data + mocked SMS/weather/payment/accounting providers

The jobs table is the central hub. Every claim, schedule, photo, document, invoice, notification, daily log, warranty, storm attribution, and referral should connect back to a job record.

6. Role model and access rules

The demo should include a role switcher so Salazar can see that owners, office staff, sales reps, crew managers, canvassers, and customers see different screens and data.

Role	Tier	Scope	Key access
Owner / Admin	1	All company records.	all_modules_full_access, manage_users, manage_settings, view_financials
Office Manager	2	All active company records except restricted admin settings.	crm_full, jobs_full, claims_full, schedule_full

Role	Tier	Scope	Key access
Production Manager	2	Production-related records and job details. Limited financial summary only.	jobs_production_full, crew_full, materials_full, punch_list_full
Insurance Coordinator	3	Claims and related jobs/documents only. No payroll or crew performance access.	claims_full, documents_claims, payments_claim_view, portal_claim_docs
Sales Rep / Estimator	3	Assigned leads, customers, inspections, jobs, and basic claim summary for assigned jobs.	crm_assigned, inspections_full_assigned, estimates_full_assigned, portal_assigned_customer_updates
Crew Manager	3	Jobs explicitly granted through crew_manager_job_access and crews assigned through crew_manager_assignments.	view_assigned_jobs, manage_assigned_crew, submit_daily_logs, complete_safety_checklists
Crew Lead	4	Assigned schedules only.	view_schedule, upload_photos, submit_field_issue, complete_checklist
Crew Member	5	Own assigned work items only.	view_today_job, acknowledge_safety, upload_limited_photos
Marketing Manager	3	StormLoop records, referrals, leads from marketing, and opted-in customer lists.	stormloop_full, referrals_full, campaigns_manage, canvassing_manage
Canvasser / Field Marketer	4	Assigned route and leads they create. No job financials or claim information.	view_assigned_route, create_door_knock, create_lead, book_inspection
Accountant	3	Finance records and job financial summaries. No crew performance reviews unless explicitly granted.	invoices_full, payments_full, job_costs_full, quickbooks_sync
Customer / Homeowner	6	Single job/property accessed through valid portal token or customer login.	portal_view_own_job, portal_message, portal_pay_invoice, portal_referral_link

6.1 Permission matrix summary

Module	Owner	Office	Sales	Insurance	Production	Crew mgr	Canvasser	Customer
CRM	Full	Full	Assigned	View claim jobs	View jobs	Limited assigned job	Create only	Own portal
Jobs	Full	Full	Assigned	Claim jobs	Production full	Assigned field only	No	Own job status
Claims	Full	Full	Assigned summary	Full	Limited production readiness	No	No	Limited customer view
CrewDesk	Full	View/approve	No	No	Full	Assigned full	No	No
StormLoop	Full	View	Leads assigned	No	No	No	Assigned route	No
Referrals	Full	Full	View assigned	No	No	No	Capture source	Own referrals
Finance	Full	Full	Limited assigned	Claim payments	Limited budget	No	No	Own invoice

7. Module requirements

CRM Command Center

Capture, qualify, assign, and convert leads into inspections and jobs.

Item	Details
Primary users	owner_admin, office_manager, sales_rep_estimator, marketing_manager
Features	Lead inbox, Customer profile, Property profile, Sales pipeline, Follow-up tasks, Lead source attribution, Lost reason tracking, Referral/storm source tags
Desktop screens	Lead Inbox, Customer Detail, Property Detail, Sales Pipeline Board, Follow-up Queue
Mobile screens	Today, Quick Lead, Customer Lookup, Inspection Booking
Data entities	customers, properties, leads, jobs, appointments, activity_logs
Acceptance criteria	Create lead in under 60 seconds from desktop or mobile.; Assign lead to rep and schedule inspection.; Convert lead to job while preserving source, storm event, and referral attribution.

Inspection App

Give reps a mobile-first inspection workflow with structured checklists, tagged photos, and inspection reports.

Item	Details
Primary users	sales_rep_estimator, office_manager
Features	Inspection checklist, Photo capture and tagging, Damage annotations, Measurement report upload, Customer signature, Inspection summary PDF, Estimate handoff
Desktop screens	Inspection Review, Photo Gallery, Documents
Mobile screens	Inspection Checklist, Camera Capture, Photo Tags, Customer Signature
Data entities	inspections, inspection_checklist_items, photos, documents, estimate_versions
Acceptance criteria	Inspection can be completed from a phone.; Photos attach to the correct job/property and support tags.; Inspection completion triggers customer update and office task.

ClaimDesk Insurance-Support Module

Track claim-related paperwork and payment milestones while keeping contractor language compliant.

Item	Details
Primary users	office_manager, insurance_coordinator, owner_admin, sales_rep_estimator
Features	Carrier and claim number, Date of loss, Adjuster contact, Adjuster appointment, Carrier scope upload, Estimate versions, Supplement tracker, ACV/RCV/deductible/payment tracker, Claim timeline, Compliance disclaimer
Desktop screens	Claims Dashboard, Claim Detail, Payment Tracker, Supplement Tracker, Missing Documents
Mobile screens	Claim Summary, Adjuster Appointment, Upload Claim Doc
Data entities	insurance_claims, claim_contacts, estimate_versions, supplements, claim_payments, documents, approved_scripts
Acceptance criteria	User can view all claim details for a job in one tab.; ACV, RCV, deductible, supplement, and final balance are tracked as separate records.; Risky phrases like claim negotiation and deductible waiver are blocked in templates.

Production Board

Move contracted roofing/remodeling jobs through materials, scheduling, crew assignment, QC, invoice, and warranty.

Item	Details
Primary users	production_manager, office_manager, owner_admin
Features	Roofing board, Remodeling board, Job status changes, Material readiness, Crew assignment, Weather delay status, Final QC, Warranty closeout
Desktop screens	Roofing Production Board, Remodeling Production Board, Production Calendar, Material Readiness Queue
Mobile screens	Production Job Summary, Issue Log
Data entities	jobs, job_schedules, material_orders, punch_list_items, warranties, activity_logs
Acceptance criteria	Job status changes are logged.; Production manager can see jobs stuck by stage.; Status changes can trigger customer notifications.

CrewDesk Operations

Give crew managers controlled field authority for assigned crews and jobs without exposing financial/claim data.

Item	Details
Primary users	crew_manager, crew_lead, crew_member, production_manager
Features	Assigned crew schedule, GPS clock-in/out, Safety checklist, Daily job log, Progress percent, Progress photos, Material request, Punch list, Certification alerts, Crew performance review
Desktop screens	Crew Dashboard, Crew Schedule, Certification Alerts, Material Approval Queue
Mobile screens	My Crew, Today Job, Safety Checklist, Daily Log, Material Request, Progress Photos
Data entities	crews, crew_members, crew_manager_assignments, crew_manager_job_access, daily_job_logs, safety_checklists, material_orders, material_order_items, crew_certifications, crew_performance_reviews
Acceptance criteria	Crew manager sees only assigned jobs and crews.; Daily job log can trigger a customer progress notification if approved.; Safety checklist must pass before job is moved to in_progress.

Customer Portal

Reduce check-in calls by giving homeowners a secure magic-link portal for job status, next steps, docs, invoices, warranty, and referrals.

Item	Details
Primary users	customer, office_manager, sales_rep_estimator
Features	Magic link access, Project timeline, Current status, Next step, Shared photos, Shared documents, Invoice/payment link, Warranty docs, Message thread, Referral link and QR
Desktop screens	Portal Preview, Portal Share Controls
Mobile screens	Customer Portal Home, Timeline, Documents, Invoice, Referral Wallet
Data entities	portal_tokens, notifications, messages, documents, photos, invoices, referrals
Acceptance criteria	Customer can open portal without app download.; Portal view is limited to one job/token.; Portal views are audited with timestamp and IP/device metadata.

Notification Engine

Centralize SMS/email/portal updates with consent tracking, approved templates, delivery logs, and automation triggers.

Item	Details
Primary users	office_manager, owner_admin, marketing_manager
Features	Transactional templates, Marketing templates, SMS consent capture, STOP/HELP handling, Delivery status, Internal alerts, Approval rules, Template variables
Desktop screens	Template Manager, Message Audit Log, Consent Center, Campaigns
Mobile screens	Send Approved Update, Customer Reply Inbox
Data entities	notification_templates, notifications, sms_consent, message_campaigns, message_audit_logs, approved_scripts
Acceptance criteria	No marketing SMS sends without marketing consent.; Transactional messages include brand and opt-out language where needed.; Each outbound/inbound message is logged.

StormLoop Growth Engine

Turn severe weather into organized, ethical, measurable sales activity: affected ZIPs, missions, routes, door knocks, leads, referrals, and revenue attribution.

Item	Details
Primary users	marketing_manager, canvasser, sales_rep_estimator, owner_admin
Features	Weather alert ingest, Affected ZIP/ZCTA impact scoring, Past customer impact list, Storm mission builder, Canvassing route assignment, Door knock outcomes, Lead capture, Do-not-knock list, Storm-to-revenue attribution
Desktop screens	Storm Dashboard, Storm Event Detail, Mission Builder, Canvassing Manager, Storm Revenue Report
Mobile screens	My Mission, Route Map, Door Pin, Quick Lead, Book Inspection
Data entities	service_areas, storm_events, storm_zip_impacts, storm_customer_impacts, marketing_missions, canvassing_routes, door_knocks, do_not_knock_addresses
Acceptance criteria	Simulated or live weather event creates affected ZIP records.; Marketing manager can launch a mission and assign routes.; Canvasser can create a lead from a door knock with consent status and storm attribution.

Referral Rewards Engine

Track referrals from customer share link to job completion and reward approval, creating long-term customer loyalty.

Item	Details
Primary users	customer, marketing_manager, office_manager, owner_admin
Features	Customer referral link, Referral QR code, Referral status tracking, Reward approval workflow, Loyalty tiers, VIP benefits, Top referrers report
Desktop screens	Referral Dashboard, Reward Approval Queue, Loyalty Accounts
Mobile screens	Portal Referral Wallet, Share Referral Link
Data entities	referrals, referral_rewards, customer_loyalty_accounts, loyalty_ledger
Acceptance criteria	Referral source follows lead through job and payment.; Reward remains pending until job is complete and paid.; Review incentives cannot require positive sentiment.

Finance, Job Costing, and Reports

Track invoices, claim payments, job costs, QuickBooks sync, revenue by source, and operational KPIs.

Item	Details
Primary users	accountant, owner_admin, office_manager
Features	Invoice tracker, Payments, Claim check tracker, Job costing, QuickBooks sync, Lead source ROI, Storm revenue attribution, Crew productivity, A/R report
Desktop screens	Reports Dashboard, Finance Dashboard, Job Cost Detail, QuickBooks Sync Queue
Mobile screens	Payment Status Summary
Data entities	invoices, payments, claim_payments, job_costs, activity_logs
Acceptance criteria	Revenue can be reported by job type, storm event, ZIP, rep, and referral source.; QuickBooks integration can be mocked in demo and enabled later.

Remodeling Expansion

Support longer remodel jobs with selections, allowances, permits, change orders, subs, daily logs, client approvals, and punch lists.

Item	Details
Primary users	production_manager, office_manager, sales_rep_estimator, customer
Features	Selections, Allowances, Change orders, Permit tracker, Subcontractor schedule, Customer approvals, Punch lists, Daily logs
Desktop screens	Remodeling Board, Selections, Change Orders, Permits, Punch List
Mobile screens	Remodeling Job Summary, Customer Approval, Punch List Photos
Data entities	remodeling_selections, change_orders, permits, punch_list_items, daily_job_logs
Acceptance criteria	Remodeling jobs use a separate pipeline from roofing.; Change orders require approval and preserve audit history.

8. StormLoop - special issue to solve

Do not call it a neighborhood stalking app. Use professional language: Neighborhood Opportunity Map, Canvassing Mission Map, Storm Opportunity Map, or StormLoop.

StormLoop solves the post-storm chaos problem: roofers manually hear about hail, guess which neighborhoods to canvass, lose leads in text messages, and cannot tell which storm or rep created revenue. StormLoop makes this measurable and repeatable.

Feature	Requirement
Weather monitor	Ingest or simulate NWS severe thunderstorm/tornado/wind/hail alerts for configured service areas.
ZIP/ZCTA impact	Intersect alert/seeded storm geometry with marketing ZIP/ZCTA areas and compute impact score. Label as approximate marketing area.
Opportunity score	Rank ZIPs by hail size, wind speed, recency, past customers, open leads, and canvassing saturation.
Mission builder	Create mission with target ZIPs, reps/routes, goals, scripts, start/end times.
Canvasser mobile	Assigned route, door script, property pin, outcome, consent, appointment, referral source.
Attribution	Lead and job preserve storm_event_id, route_id, canvasser, ZIP, and referral_id.
Compliance	Do-not-knock list, no-soliciting notes, approved scripts, right-to-cancel materials, consent capture.

8.1 Storm severity thresholds

Use these default thresholds for demo scoring: base severe = 1.00 inch hail or 58 mph wind; considerable = 1.75 inch hail or 70 mph wind; destructive = 2.75 inch hail or 80 mph wind. See sources S2 and S3.

9. Insurance-support and compliance

The system should help Salazar organize claim-related construction documents, not act as a public adjuster. Every claim-related screen and template should use careful wording.

Area	Rule	Source
Insurance language	Use insurance-support documentation wording. Do not say the contractor files, negotiates, adjusts, or settles claims for homeowners.	S1
Deductibles	Block language that waives, rebates, absorbs, discounts, or offsets insurance deductibles. Rewards must not reduce deductible obligations.	S1
Door-to-door canvassing	Support approved scripts, rep ID, do-not-knock list, no high-pressure language, and right-to-cancel form workflow when contracts are signed at a home.	S12
SMS consent	Store transactional and marketing consent separately. Respect STOP/HELP and never send marketing SMS without documented consent.	S5/S6
Reviews and incentives	Do not condition rewards on positive reviews. Use honest review language and disclose incentives where applicable.	S7
Storm impact claims	Say affected marketing ZIP/ZCTA or elevated inspection opportunity. Do not claim a specific home is damaged without inspection.	S2/S3/S13
Privacy and access	Apply RLS/role-based access. Crew/canvassers should not see financial or insurance details they do not need.	Internal design

9.1 Blocked language examples

- Do not say: We file your claim.
- Do not say: We negotiate with your carrier.
- Do not say: We represent you against the insurance company.
- Do not say: We waive or absorb your deductible.
- Do not say: Free roof paid by insurance.
- Preferred: We provide contractor estimates, photos, and technical documentation; the homeowner owns the insurance claim.

10. Database schema summary

The JSON file contains the full machine-readable schema with fields, relationships, indexes, and RLS scope. The PDF summarizes the entity groups for build planning.

Entity	Module	Purpose	Key fields
companies	core	Tenant/company record for single or multi-company support.	id:uuid PK, name:varchar(180), legal_name:varchar(180), phone:varchar(30), email:varchar(255), website:varchar(255), primary_address:text, timezone:varchar(60)
users	identity	Internal staff, crew, canvassers, and external limited-access users.	id:uuid PK, company_id:uuid FK, first_name:varchar(80), last_name:varchar(80), email:varchar(255), phone:varchar(30), role:user_role enum, is_active:boolean

Entity	Module	Purpose	Key fields
role_permissions	identity	Permission matrix for module/action/data-scope control.	id:uuid PK, company_id:uuid FK, role:user_role enum, module:varchar(80), action:varchar(80), scope:varchar(80), is_allowed:boolean, created_at:timestampz
customers	crm	Homeowners/customers and referrers.	id:uuid PK, company_id:uuid FK, first_name:varchar(80), last_name:varchar(80), email:varchar(255), phone:varchar(30), preferred_language:varchar(20), sms_transactional_opt_in:boolean
properties	crm	Physical service locations; one customer can have multiple properties.	id:uuid PK, company_id:uuid FK, customer_id:uuid FK, address_line1:varchar(180), address_line2:varchar(80), city:varchar(100), state:char(2), zip_code:varchar(10)
leads	crm	Pre-job opportunities from web, phone, door knock, storm, referral, repeat customer, or yard sign.	id:uuid PK, company_id:uuid FK, customer_id:uuid FK, property_id:uuid FK, assigned_rep_id:uuid FK, source:lead_source enum, source_detail:varchar(180), storm_event_id:uuid FK
jobs	jobs	Central hub. Every claim, schedule, photo, invoice, notification, crew log, and warranty keys back to jobs.	id:uuid PK, company_id:uuid FK, customer_id:uuid FK, property_id:uuid FK, lead_id:uuid FK, assigned_rep_id:uuid FK, job_type:job_type enum, status:job_status enum
appointments	crm	Inspections, adjuster meetings, customer meetings, deliveries, final walkthroughs.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, lead_id:uuid FK, customer_id:uuid FK, appointment_type:appointment_type enum, starts_at:timestampz, ends_at:timestampz
inspections	inspection	Structured inspection record for roofing/remodeling inspections.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, lead_id:uuid FK, property_id:uuid FK, inspector_user_id:uuid FK, inspection_type:inspection_type enum, damage_found:boolean
inspection_checklist_items	inspection	Checklist answers for each inspection.	id:uuid PK, inspection_id:uuid FK, section:varchar(80), item_key:varchar(80), label:varchar(180), value:jsonb, notes:text, requires_photo:boolean
photos	documents	Tagged job/property photos: before, during, after, damage, cleanup, warranty proof.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, inspection_id:uuid FK, uploaded_by_id:uuid FK, storage_url:varchar(255), thumbnail_url:varchar(255), tags:text[]
documents	documents	Estimates, contracts, insurance scopes, PDFs, warranties, invoices, permits, reports.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, customer_id:uuid FK, doc_type:document_type enum, title:varchar(180), storage_url:varchar(255), version:integer

Entity	Module	Purpose	Key fields
insurance_claims	claimdesk	Claim-support record for insurance-related jobs.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, carrier:varchar(120), claim_number:varchar(80), policyholder_name:varchar(160), date_of_loss:date, loss_type:loss_type enum
claim_contacts	claimdesk	Adjusters, desk adjusters, claim reps, mortgage contacts, supplement partners.	id:uuid PK, claim_id:uuid FK, contact_type:claim_contact_type enum, name:varchar(160), company:varchar(160), phone:varchar(30), email:varchar(255), notes:text
estimate_versions	claimdesk	Contractor estimates, carrier scopes, Xactimate PDFs, proposal versions.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, claim_id:uuid FK, estimate_type:estimate_type enum, version:integer, amount:numeric(12,2), document_id:uuid FK
supplements	claimdesk	Supplement request tracker and outcome history.	id:uuid PK, company_id:uuid FK, claim_id:uuid FK, estimate_version_id:uuid FK, amount_requested:numeric(12,2), amount_approved:numeric(12,2), status:supplement_status enum, sent_at:timestampz
claim_payments	claimdesk_financ e	Separate check/payment milestones for ACV, deductible, RCV, supplement, upgrade, and final balance.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, claim_id:uuid FK, payment_type:claim_payment_type enum, expected_amount:numeric(12,2), received_amount:numeric(12,2), status:payment_status enum
crews	crewdesk	Crew groups for roofing/remodeling production.	id:uuid PK, company_id:uuid FK, name:varchar(120), crew_type:crew_type enum, primary_manager_id:uuid FK, is_active:boolean, created_at:timestampz
crew_members	crewdesk	Users assigned to crews plus worker metadata.	id:uuid PK, company_id:uuid FK, crew_id:uuid FK, user_id:uuid FK, worker_role:crew_worker_role enum, is_subcontractor:boolean, hourly_rate:numeric(10,2), active_from:date
crew_manager_assignments	crewdesk	Map crew managers to one or more crews; supports primary/backup managers.	id:uuid PK, company_id:uuid FK, crew_id:uuid FK, manager_user_id:uuid FK, is_primary:boolean, active_from:date, active_to:date
crew_manager_job_accesses	crewdesk	Explicit grants that control which jobs a crew manager can access and at what level.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, manager_user_id:uuid FK, access_level:crew_job_access_level enum, granted_by_id:uuid FK, expires_at:timestampz, created_at:timestampz
job_schedules	crewdesk	Crew/day assignments for jobs.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, crew_id:uuid FK, scheduled_date:date, start_time:time, end_time:time, work_scope:text

Entity	Module	Purpose	Key fields
daily_job_logs	crewdesk	End-of-day or progress logs submitted by crew manager/lead.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, schedule_id:uuid FK, submitted_by_id:uuid FK, log_date:date, summary:text, progress_percent:integer
safety_checklists	crewdesk	Daily pre-work signoff; can gate job start.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, schedule_id:uuid FK, submitted_by_id:uuid FK, ppe_checked:boolean, fall_protection_checked:boolean, ladder_checked:boolean
material_orders	crewdesk	Material requests and order tracking.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, requested_by_id:uuid FK, supplier_name:varchar(160), status:material_order_status enum, needed_by:date, approved_by_id:uuid FK
material_order_items	crewdesk	Line items for material orders/requests.	id:uuid PK, material_order_id:uuid FK, sku:varchar(80), name:varchar(180), quantity:numeric(12,2), unit:varchar(40), unit_cost:numeric(10,2), notes:text
crew_certifications	crewdesk	OSHA, fall protection, manufacturer certifications, expiration alerts.	id:uuid PK, company_id:uuid FK, user_id:uuid FK, cert_type:certification_type enum, cert_number:varchar(120), issued_at:date, expires_at:date, document_id:uuid FK
crew_performance_reviews	crewdesk	Optional job-level or periodic worker scorecards.	id:uuid PK, company_id:uuid FK, crew_member_user_id:uuid FK, reviewer_user_id:uuid FK, job_id:uuid FK, quality_score:integer, safety_score:integer, timeliness_score:integer
portal_tokens	portal	Secure magic-link access for customer portal without app download.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, customer_id:uuid FK, token_hash:varchar(255), expires_at:timestampz, last_viewed_at:timestampz, view_count:integer
notification_templates	notifications	Approved SMS/email/portal templates with variables.	id:uuid PK, company_id:uuid FK, template_key:varchar(100), channel:notification_channel enum, purpose:notification_purpose enum, subject:varchar(180), body:text, requires_approval:boolean
notifications	notifications	Outbound and inbound communication records.	id:uuid PK, company_id:uuid FK, customer_id:uuid FK, job_id:uuid FK, template_id:uuid FK, channel:notification_channel enum, direction:message_direction enum, body:text
sms_consent	compliance	Consent evidence for transactional and marketing SMS.	id:uuid PK, company_id:uuid FK, customer_id:uuid FK, phone:varchar(30), consent_type:sms_consent_type enum, consent_source:consent_source enum, consent_text:text, consented_at:timestampz

Entity	Module	Purpose	Key fields
approved_scripts	compliance	Approved canvassing scripts, insurance disclaimers, deductible warnings, SMS snippets, review/referral language.	id:uuid PK, company_id:uuid FK, script_type:script_type enum, title:varchar(180), body:text, blocked_terms:text[], approved_by_id:uuid FK, approved_at:timestampz
invoices	finance	Customer invoices and payment status.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, customer_id:uuid FK, invoice_number:varchar(80), amount:numeric(12,2), balance_due:numeric(12,2), status:invoice_status enum
payments	finance	Non-claim payment transactions, payment links, card/ACH/check status.	id:uuid PK, company_id:uuid FK, invoice_id:uuid FK, job_id:uuid FK, payment_method:payment_method enum, amount:numeric(12,2), status:payment_status enum, processor_payment_id:varchar(180)
job_costs	finance	Cost tracking for labor, materials, subcontractors, permits, overhead, and gross margin.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, cost_type:job_cost_type enum, description:varchar(180), estimated_amount:numeric(12,2), actual_amount:numeric(12,2), vendor:varchar(160)
service_areas	stormloop	Configured cities, ZIPs/ZCTAs, counties, and radius markets monitored for weather events.	id:uuid PK, company_id:uuid FK, name:varchar(160), zip_codes:text[], counties:text[], center_lat:numeric(10,7), center_lng:numeric(10,7), radius_miles:integer
storm_events	stormloop	Detected or simulated hail/wind/tornado/severe storm events.	id:uuid PK, company_id:uuid FK, event_name:varchar(180), event_type:storm_event_type enum, source:weather_source enum, started_at:timestampz, ended_at:timestampz, max_hail_size_in:numeric(4,2)
storm_zip_impacts	stormloop	Impact ranking by affected marketing ZIP/ZCTA.	id:uuid PK, storm_event_id:uuid FK, zip_code:varchar(10), city:varchar(100), county:varchar(100), hail_size_estimate_in:numeric(4,2), wind_speed_estimate_mph:integer, impact_score:integer
storm_customer_impacts	stormloop	Past/existing customers located in a storm impact area for check-in campaigns.	id:uuid PK, storm_event_id:uuid FK, customer_id:uuid FK, property_id:uuid FK, zip_code:varchar(10), last_job_id:uuid FK, recommended_action:varchar(120), contacted_at:timestampz
marketing_missions	stormloop	Canvassing/customer-check-in missions launched from storm events.	id:uuid PK, company_id:uuid FK, storm_event_id:uuid FK, mission_name:varchar(180), mission_type:mission_type enum, target_zip_codes:text[], goal_doors:integer, goal_inspections:integer

Entity	Module	Purpose	Key fields
canvassing_routes	stormloop	Territory/route assigned to a canvasser.	id:uuid PK, mission_id:uuid FK, assigned_user_id:uuid FK, route_name:varchar(120), zip_code:varchar(10), territory_polygon:geometry, status:route_status enum, doors_targeted:integer
door_knocks	stormloop	Property-level canvassing interaction log and lead creation source.	id:uuid PK, route_id:uuid FK, property_address:varchar(255), lat:numeric(10,7), lng:numeric(10,7), status:door_status enum, homeowner_name:varchar(160), phone:varchar(30)
do_not_knock_addresses	compliance	Addresses where canvassers should not knock or solicit.	id:uuid PK, company_id:uuid FK, address_line1:varchar(180), city:varchar(100), state:char(2), zip_code:varchar(10), reason:varchar(180), source:varchar(80)
referrals	referrals	Tracks referred customers from link/QR/manual entry through lead, job, payment, reward.	id:uuid PK, company_id:uuid FK, referrer_customer_id:uuid FK, referred_customer_id:uuid FK, lead_id:uuid FK, job_id:uuid FK, referral_code:varchar(40), status:referral_status enum
referral_rewards	referrals	Reward approval and issue tracking.	id:uuid PK, referral_id:uuid FK, reward_type:reward_type enum, reward_amount:numeric(12,2), reward_description:varchar(255), approved_by_id:uuid FK, approved_at:timestampz, issued_at:timestampz
customer_loyalty_accounts	referrals	Lifetime referral wallet and VIP status for customers.	id:uuid PK, company_id:uuid FK, customer_id:uuid FK, tier:loyalty_tier enum, points:integer, lifetime_referrals:integer, lifetime_revenue_referred:numeric(12,2), vip_status:boolean
loyalty_ledger	referrals	Points/reward ledger events.	id:uuid PK, loyalty_account_id:uuid FK, event_type:loyalty_event_type enum, points_change:integer, description:varchar(255), referral_id:uuid FK, created_at:timestampz
remodeling_selections	remodeling	Selections and allowance tracking for remodeling jobs.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, category:varchar(80), item_name:varchar(180), allowance_amount:numeric(12,2), selected_vendor:varchar(180), selected_option:varchar(180)
change_orders	remodeling	Cost/schedule changes needing approval.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, title:varchar(180), description:text, cost_change:numeric(12,2), schedule_change_days:integer, status:change_order_status enum

Entity	Module	Purpose	Key fields
permits	remodeling_production	Permit and inspection tracker.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, permit_type:varchar(80), jurisdiction:varchar(160), permit_number:varchar(120), status:permit_status enum, submitted_at:date
punch_list_items	production	Final punch list and cleanup/QC tasks.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, title:varchar(180), description:text, assigned_to_id:uuid FK, status:punch_status enum, photo_required:boolean
warranties	production	Workmanship/manufacture warranty records and portal docs.	id:uuid PK, company_id:uuid FK, job_id:uuid FK, warranty_type:warranty_type enum, provider:varchar(160), start_date:date, end_date:date, document_id:uuid FK
activity_logs	audit	Immutable timeline/audit log for important actions.	id:uuid PK, company_id:uuid FK, actor_user_id:uuid FK, customer_id:uuid FK, job_id:uuid FK, entity_name:varchar(120), entity_id:uuid, action:varchar(120)

11. API endpoint plan

REST JSON is sufficient for the demo. The JSON file can be converted to OpenAPI later. All endpoints must enforce company_id and role-based scope.

Method	Path	Module	Description
GET	/api/v1/me	identity	Return current user, role, company, and permissions.
GET	/api/v1/dashboard/owner	dashboard	Owner KPI cards, stuck jobs, claims needing attention, crew status, storm alerts, and referral revenue.
GET	/api/v1/leads	crm	List leads filtered by status, source, assigned rep, storm, referral.
POST	/api/v1/leads	crm	Create lead; supports storm_event_id and referral_id attribution.
PATCH	/api/v1/leads/{lead_id}	crm	Update lead status, assignment, follow-up, lost reason.
POST	/api/v1/leads/{lead_id}/convert-to-job	crm	Convert lead into job while preserving attribution.
GET	/api/v1/jobs	jobs	List jobs by board/status/job_type/assigned user.
GET	/api/v1/jobs/{job_id}	jobs	Read job detail respecting RLS scope.
PATCH	/api/v1/jobs/{job_id}/status	jobs	Update job stage; may trigger notification and audit log.
POST	/api/v1/jobs/{job_id}/photos	documents	Upload/tag job photos.
GET	/api/v1/claims	claimdesk	Claims dashboard with filters and missing docs.
POST	/api/v1/jobs/{job_id}/claim	claimdesk	Create claim-support record for job.
PATCH	/api/v1/claims/{claim_id}	claimdesk	Update carrier, claim number, status, adjuster, amounts, compliance acknowledgement.
POST	/api/v1/claims/{claim_id}/payments	claimdesk	Add ACV, RCV, deductible, supplement, or final payment milestone.
POST	/api/v1/claims/{claim_id}/supplements	claimdesk	Create supplement tracker entry.
GET	/api/v1/crews	crewdesk	List crews with role scope.

Method	Path	Module	Description
POST	/api/v1/jobs/{job_id}/schedule	crewdesk	Assign crew/date/scope to job.
POST	/api/v1/schedules/{schedule_id}/safety-checklist	crewdesk	Submit safety checklist. Can gate job start.
POST	/api/v1/jobs/{job_id}/daily-log	crewdesk	Submit daily progress log; optional customer notification.
POST	/api/v1/jobs/{job_id}/material-requests	crewdesk	Request missing materials from field.
PATCH	/api/v1/material-requests/{order_id}/approve	crewdesk	Approve/reject material request.
GET	/api/v1/portal/{token}	portal	Return customer-safe portal view for one job.
POST	/api/v1/portal/{token}/messages	portal	Customer sends message to office.
POST	/api/v1/notifications/send	notifications	Send approved template via SMS/email/portal after consent checks.
GET	/api/v1/consents/{customer_id}	compliance	Read SMS consent/opt-out history.
POST	/api/v1/storms/simulate	stormloop	Demo endpoint to create a simulated storm alert and ZIP impacts.
GET	/api/v1/storms	stormloop	List storm events and severity.
GET	/api/v1/storms/{storm_id}/impacts	stormloop	Return ZIP impacts, past customers, open leads, and opportunity scores.
POST	/api/v1/missions	stormloop	Create canvassing/customer check-in mission.
POST	/api/v1/missions/{mission_id}/routes	stormloop	Assign route to canvasser.
POST	/api/v1/routes/{route_id}/door-knocks	stormloop	Log door knock and optionally create a lead/appointment.
GET	/api/v1/referrals	referrals	Referral dashboard and statuses.
POST	/api/v1/referrals	referrals	Create referral from portal link, QR, or manual entry.
PATCH	/api/v1/referral-rewards/{reward_id}/approve	referrals	Approve referral reward after completion/payment checks.
POST	/api/v1/dev/simulate/status-change	demo	Demo-only: change status and generate sample notification/audit record.

12. Automations and notifications

12.1 Automation triggers

Automation	Trigger	Actions
lead_assignment	lead.created	assign default rep or queue owner; create follow-up task; log activity
inspection_reminder	appointment.starts_at - 24h	send appointment_confirmed/reminder
claim_missing_scope	claim.status=claim_number_received for 7 days	create office task; show on claims needing attention
safety_gate	job.schedule.start_day	block job status move to in_progress; alert production manager
daily_log_customer_update	daily_job_log.created	send daily_progress template; update portal timeline; log audit
material_request_approval	material_order.created	notify office/production; add to material approval queue
cert_expiry_warning	daily 8am	alert production manager; show certification badge
storm_alert_ingest	weather_worker.alert_received	create storm_event; calculate storm_zip_impacts; create storm_customer_impacts; notify marketing manager
mission_created	marketing_mission.created	assign canvasser routes; notify canvassers; add mission KPIs

Automation	Trigger	Actions
door_knock_to_lead	door_knock.outcome=wants_inspection	create customer/property/lead; book appointment if selected; preserve route/storm/referral attribution
referral_reward_pending	referred job status=paid_closed	mark reward pending approval; notify marketing manager; update loyalty ledger pending
warranty_review_referral	warranty.created	send warranty_uploaded; send review_request; show referral link in portal

12.2 Notification templates

Template	Purpose	Trigger	Message body
appointment_confirmed	transactional	appointment.created	Salazar Roofing & Remodeling: Your {appointment_type} is scheduled for {appointment_datetime}. Reply STOP to opt out.
inspection_complete	transactional	inspection.completed	Salazar Roofing & Remodeling: Your inspection is complete. We uploaded your inspection summary to your project portal: {portal_link}. Reply STOP to opt out.
claim_docs_received	transactional	claim.scope_uploaded	Salazar Roofing & Remodeling: We received your insurance scope/document. Our team is reviewing the construction details and will update your portal with next steps.
install_scheduled	transactional	job.status.install_scheduled	Salazar Roofing & Remodeling: Your roof installation is scheduled for {install_date}. Materials are planned for {delivery_date}. Weather may affect timing.
crew_on_site	transactional	crew.clocked_in	Salazar Roofing & Remodeling: Our crew has arrived and started your project. Progress photos will be added to your portal.
daily_progress	transactional	daily_job_log.notify_customer_true	Salazar Roofing & Remodeling: Today progress update: {progress_summary}. Your project is {progress_percent}% complete. View details: {portal_link}.
weather_delay	transactional	job.weather_delay	Salazar Roofing & Remodeling: Weather has delayed today work. We will update your schedule as soon as conditions allow. View updates: {portal_link}.
warranty_uploaded	transactional	warranty.created	Salazar Roofing & Remodeling: Your warranty documents are now available in your portal. Thank you for trusting us with your home.
storm_check_in	marketing_or_existing_customer_check_in	storm_customer_impact.approved_campaign	Salazar Roofing & Remodeling: We saw severe weather reported near your area. We hope your home is safe. Reply INSPECT if you want a no-pressure roof check. Reply STOP to opt out.
referral_status_update	transactional	referral.status_changed	Salazar Roofing & Remodeling: Your referral status changed to {referral_status}. Track rewards here: {portal_link}.
review_request	transactional_review_request	job.closed	Salazar Roofing & Remodeling: Thank you for choosing us. Share an honest review here: {review_link}. Your feedback helps local homeowners.

13. Demo user story and script

Use one coherent story rather than jumping between unrelated screens. Maria Chen is the anchor job for the demo.

- Start on Owner Dashboard. Show active leads, production jobs, claims needing attention, crews on-site, and active StormLoop alert.
- Open StormLoop. Show Rockwall-Fate storm event, affected ZIPs 75087, 75189, 75132, and opportunity scores.
- Launch Mission. Assign canvasser route and show goal doors/inspections/contracts.

- Switch to canvasser mobile route. Log a door knock and create a lead with consent and inspection appointment.
- Open Maria Chen job. Show overview, contract amount, progress percent, and source attribution.
- Open ClaimDesk tab. Show carrier, claim number, adjuster, date of loss, ACV received, RCV pending, supplement submitted, deductible tracked.
- Open Crew tab. Show Crew A assigned, safety checklist passed, daily log, progress photos, material request, certification alert.
- Open Notifications/Portal tab. Show SMS history, magic link, portal views, timeline, documents, photos, invoice, warranty placeholder.
- Open Referral Wallet. Show Maria referral link/QR and a neighbor referral status moving from submitted to inspection scheduled.
- Close on Reports. Show revenue by storm, lead source, referral, ZIP, and rep.

13.1 Seed data to load

Data group	Seed values
KPI cards	Active leads: 18; In production: 12; Claims needing attention: 3; Crew on-site today: 7
Storm event	Rockwall-Fate Hail/Wind Event - 75087 score 92, 75189 score 76, 75132 score 64
Anchor customer	Maria Chen - 4521 Ridgecrest Dr, Rockwall, TX 75087 - SF-2026-089234
Crews	Crew A on Maria Chen roof replacement; Crew B on Sandra Kim remodeling

14. Build backlog and definition of done

Foundation

- Create app shell/navigation
- Implement auth role switcher for demo
- Create database migrations for core entities
- Load seed data
- Create owner dashboard cards

CRM and Jobs

- Lead inbox
- Jobs board
- Job detail overview
- Activity timeline
- Document/photo mock upload

ClaimDesk

- Insurance tab
- Claim payment cards
- Adjuster panel
- Supplement tracker
- Compliance disclaimer

CrewDesk

- Crew dashboard
- Crew manager role scope
- Safety checklist
- Daily log
- Material request approval

Customer Portal

- Magic link token route
- Portal timeline
- Shared docs/photos
- Invoice status
- Referral wallet

StormLoop

- Storm event dashboard
- Affected ZIP impact table
- Mission builder
- Canvasser mobile route
- Door knock to lead

Referrals and Reports

- Referral dashboard
- Reward queue
- Revenue by source
- Storm attribution report

Demo Polish

- Guided demo script path
- Clickable Maria Chen story
- Role-switcher
- Mock notification log
- PDF/JSON handoff loaded into repo docs

Definition of done

- User can complete the demo story from storm alert to lead, inspection, claim, crew update, portal view, and referral.
- Role switcher proves crew/canvasser/customer cannot see restricted financial/claim data.
- All KPI cards and dashboards use seeded data.
- Status changes create activity log and mock notification records.
- No production SMS, weather, accounting, or payment calls are required for demo; all external providers can be mocked.

15. Integration roadmap

Integration	Phase	Purpose	Notes
National Weather Service Alerts API	MVP	Storm alert ingest for service areas.	Poll conservatively; use simulated alert endpoint for demo.
NOAA/NCEI Storm Events Database	Phase 2	Historical verification and reporting.	Not instant; better for historical records than same-day canvassing.
ZCTA boundary data	MVP	Approximate ZIP/ZCTA mapping from alert polygons.	Label outputs as marketing ZIP/ZCTA areas, not guaranteed USPS ZIP impact.
Twilio or messaging provider	MVP mock, Phase 2 live	SMS notifications, replies, STOP/HELP handling.	A2P 10DLC registration and consent evidence required before production marketing SMS.
SendGrid/Postmark	MVP mock, Phase 2 live	Email notifications and portal link delivery.	Use transactional domain auth before production.
QuickBooks Online	Phase 2	Invoices, payments, job costs, project profitability.	MVP can export CSV or mock sync queue.
CompanyCam	Phase 2	Photo documentation sync.	MVP should provide native photo upload/tagging.
EagleView / HOVER / Roofr reports	Phase 2	Measurement report storage and estimate inputs.	Do not rebuild roof measurements in MVP.
Xactimate / ESX files	Phase 3	Carrier/contractor estimate records.	MVP stores PDFs/files and estimate version metadata.
Supplier APIs (ABC/SRS/QXO)	Phase 3	Material pricing/order delivery status.	MVP material requests and PDF uploads are enough.

16. Dev team checklist

- Create migrations for entities in JSON database_schema.entities.
- Implement role switcher and enforce scoped mock data at UI/API layer for demo.
- Seed the company, users, customers, storm event, ZIP impacts, jobs, claim payments, crews, notifications, portal token, and referral records.
- Build owner dashboard, jobs board, job detail, ClaimDesk, CrewDesk, Customer Portal, StormLoop, Mission Builder, Canvasser Mobile, Referral Dashboard, and Reports.
- Mock external providers: weather, SMS, email, payments, QuickBooks, file storage.
- Show audit trail for status changes, portal views, messages, claim updates, and material approvals.
- Add compliance copy to insurance, canvassing, SMS consent, referral/review, and deductible-related screens.
- Keep demo data clearly marked as sample/demo only.

17. Research and compliance references

ID	Source	Used for	URL
S1	Texas Department of Insurance - Roofing and insurance: Know the law	Texas insurance-claim language, public adjuster prohibition, deductible-waiver guardrails.	https://www.tdi.texas.gov/consumer/storms/roofing-and-insurance-know-the-law.html
S2	National Weather Service - Alerts API documentation	Storm alert integration design and polling guidance.	https://www.weather.gov/documentation/services-web-alerts
S3	National Weather Service - Severe Thunderstorm Warning damage threat categories	StormLoop severity thresholds: 1.00 in hail/58 mph wind; 1.75 in/70 mph; 2.75 in/80 mph.	https://www.weather.gov/lot/SevereThunderstormWarningsUpdate
S4	NOAA NCEI - Storm Events Database	Historical storm event verification and later reporting.	https://www.ncei.noaa.gov/stormevents/

ID	Source	Used for	URL
S5	Twilio - A2P 10DLC campaign approval best practices	SMS consent, opt-in, opt-out, HELP/STOP, campaign registration requirements.	https://help.twilio.com/articles/11847054539547-A2P-10DLC-Campaign-Approval-Best-Practices
S6	Twilio - How to collect customer consent in A2P 10DLC campaigns	Consent capture design for transactional vs marketing messages.	https://help.twilio.com/articles/26147853183259-How-to-collect-customer-consent-in-A2P-10DLC-campaigns
S7	Federal Trade Commission - Consumer Reviews and Testimonials Rule Q&A	Referral/review incentive rules and anti-fake-review guardrails.	https://www.ftc.gov/business-guidance/resources/consumer-reviews-testimonials-rule-questions-answers
S8	AccuLynx - Roofing customer portal software	Customer portal feature validation: status, appointments, documents, billing, payments, messages.	https://acculynx.com/features/customer-portal-software/
S9	AccuLynx - Plan options	Market baseline for roofing CRM pricing and core features.	https://acculynx.com/plan-options/
S10	JobNimbus - Pricing and packages	Competitor packaging, user tiers, boards, automations, mobile app, integrations.	https://www.jobnimbus.com/pricing/
S11	QuickBooks - Job and project costing software	Accounting/job-costing integration requirements.	https://quickbooks.intuit.com/accounting/job-costing/
S12	Texas Attorney General - Door-to-door sales and 3-day right of rescission	Canvassing/door-to-door compliance requirements.	https://www.texasattorneygeneral.gov/consumer-protection/home-real-estate-and-travel/door-to-door-sales-3-day-right-rescission
S13	Data.gov/Census - ZCTA boundaries are approximate ZIP representations	ZIP/ZCTA caveat for StormLoop affected-area mapping.	https://catalog.data.gov/dataset/tiger-line-shapefile-2019-2010-nation-u-s-2010-census-5-digit-zip-code-tabulation-area-zcta5-na

Note: This document is a product/design handoff, not legal advice. Legal/compliance language should be reviewed by qualified counsel before production use, especially insurance-related, door-to-door, SMS marketing, and referral/review incentive workflows.